

ParaBank Automation - Test Plan

Project Overview:

This project aims to automate the testing of the ParaBank banking application using Selenium WebDriver, Java, TestNG, Maven, Jenkins, and GitHub. The automation framework is developed using the Page Object Model (POM) design pattern.

Objective:

The objective of this project is to automate important banking functionalities and reduce manual testing effort.

Modules Covered:

- User Registration
- Login
- Open Account
- Account Overview
- Transfer Funds
- Bill Pay

Tools used:

- Java
- Selenium WebDriver
- TestNG
- Maven
- Jenkins
- Github

Test Approach:

The framework follows the Page Object Model (POM) design pattern. Test scripts are created for positive test scenarios and known defect scenarios are maintained separately.



Test Environment:

- Operating System: macOS
- Browser: Chrome, Firefox
- Application URL: <https://parabank.parasoft.com/parabank/index.htm>

Test Scenarios:

Registration:

- Register a new user with valid details.

Login:

- Login with valid credentials.

Open Account:

- Create a new savings account.

Account Overview:

- Verify account details are displayed.

Transfer Funds:

- Transfer funds between accounts.

Bill Pay:

- Make bill payments successfully.

TC ID	Module	Test Scenario	Test Steps	Expected Result	Actual Result	Status
TC_001	Registration	Register new user	Enter valid details and click Register	User registered successfully	User registered successfully	Pass
TC_002	Login	Valid Login	Enter valid username/password	User logged in	User logged in	Pass
TC_003	Login	Invalid Login	Enter invalid credentials	Error message displayed	Error message displayed	Pass
TC_004	Login	Empty Login	Leave fields blank and click Login	Validation message displayed	Validation message displayed	Pass
TC_005	Open Account	Create Savings Account	Open new account	Account created	Account created	Pass
TC_006	Account Overview	View Account Details	Navigate to Accounts Overview	Account details displayed	Account details displayed	Pass
TC_007	Transfer Funds	Valid Transfer	Transfer funds between accounts	Transfer successful	Transfer successful	Pass
TC_008	Bill Pay	Valid Bill Payment	Pay bill with valid amount	Payment successful	Payment successful	Pass

Defects:

- Same account transfer is allowed.
- Zero amount transfer is allowed.
- Zero amount bill payment is allowed.
- Negative amount bill payment is allowed.

Jenkins Integration:

Jenkins is integrated with the automation framework to execute automated test runs. The Jenkins pipeline is configured to pull the latest code from the GitHub repository, build the project using Maven, execute the TestNG test suite, and generate test reports. The build is triggered manually using the "Build Now" option in Jenkins.

The screenshot displays the Jenkins web interface for a project named "Capstone Project". The top navigation bar shows the Jenkins logo, the project name, and the build number "#15". A green checkmark icon indicates a successful build. The main content area is divided into a left sidebar with navigation links and a central panel showing build details.

Build Details:

- Build #15** (10 Jun 2026, 14:14:45)
- Status:** Successful (Green checkmark)
- Build Artifacts:** A list of files generated during the build, each with a size and a "view" link.

Artifact	Size	Action
bullet_point.png	356 B	view
collapseall.gif	157 B	view
emailable-report.html	9.49 KIB	view
failed.png	977 B	view
index.html	43.42 KIB	view
jquery-3.6.0.min.js	87.40 KIB	view
navigator-bullet.png	352 B	view
passed.png	1019 B	view
skipped.png	967 B	view
TEST-TestSuite.xml	38.68 KIB	view
testng-reports.css	5.23 KIB	view
testng-reports.js	3.53 KIB	view
testng-reports1.css	6.88 KIB	view
testng-reports2.js	3.53 KIB	view
testng-results.xml	36.51 KIB	view
TestSuite.txt	269 B	view
- Started by user:** Surya Prakash J
- This run spent:**
 - 13 ms waiting;
 - 1 min 40 sec build duration;
 - 1 min 40 sec total from scheduled to completion.
- Revision:** 57aeaf1c4e191d4c5220d02172560fdeb72234e
- Repository:** https://github.com/suryaprakashgithubj/Capstone_Project.git
- Branch:** refs/remotes/origin/main

The left sidebar contains links for "Status", "Changes", "Console Output", "Edit Build Information", "Delete build '#15'", "Timings", "Git Build Data", "Pipeline Overview", "Restart from Stage", "Replay", "Pipeline Steps", "Workspaces", and "Previous Build".

Conclusion:

The ParaBank Automation Framework successfully automates major banking functionalities. The framework supports automated execution, reporting, cross-browser testing, and Jenkins integration.

Github Repository:

https://github.com/suryaprakashgithubj/Capstone_Project.git

Defect Report

Bug_01: Same Account Transfer Allowed

Steps:

1. Login
2. Open Transfer Funds
3. Select same account as source and destination
4. Enter amount
5. Click Transfer

Expected Result:

System should not allow transfer between same accounts.

Actual Result:

Transfer is completed successfully.

BUG-002: Zero Amount Transfer Allowed

Steps:

1. Launch the ParaBank application.
2. Login with valid credentials.
3. Click Transfer Funds.
4. Enter 0 in the Amount field.
5. Select source and destination accounts.
6. Click Transfer.

Expected Result:

System should display a validation message and prevent the transfer.

Actual Result:

Transfer is completed successfully with zero amount.

BUG-003: Zero Amount Bill Payment Allowed

Steps:

1. Launch the ParaBank application.
2. Login with valid credentials.

3. Click Bill Pay.
4. Enter valid payee details.
5. Enter 0 in the Amount field.
6. Select the account.
7. Click Send Payment.

Expected Result:

System should display a validation error and prevent payment.

Actual Result:

Bill payment is processed successfully with zero amount.

BUG-004: Negative Amount Bill Payment Allowed

Steps:

1. Launch the ParaBank application.
2. Login with valid credentials.
3. Click Bill Pay.
4. Enter valid payee details.
5. Enter -100 in the Amount field.
6. Select the account.
7. Click Send Payment.

Expected Result:

System should reject negative values and display an error message.

Actual Result:

Bill payment is processed successfully with a negative amount.

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Transfer Funds allows transfer between the same source and destination account

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To Do

Improve Bug

Description

The application allows users to transfer funds when the source and destination accounts are the same. The system should prevent this action and display an appropriate validation message.

Steps to Reproduce:

1. Login to ParaBank.

2. Navigate to Transfer Funds.

3. Select the same account in both From Account and To Account.

4. Enter a valid amount.

5. Click Transfer.

Expected Result:

The system should not allow the transfer and should display an error message.

Actual Result:

The transfer is processed successfully even when the source and destination accounts are the same.

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System allows transfer of zero amount and displays successful transaction message

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To Do

Improve Bug

Description

Steps to Reproduce:

1. Login to ParaBank application.

2. Navigate to Transfer Funds.

3. Enter transfer amount as 0.

4. Select source and destination accounts.

5. Click Transfer.

Expected Result:

• System should validate the amount entered.

• Transfer amount should be greater than zero.

• Appropriate validation message should be displayed.

Actual Result:

• System processes the transaction successfully.

• "Transfer Complete!" message is displayed.

• Confirmation shows \$0.00 has been transferred.

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PARA-1 / PARA-6

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System allows bill payment with zero amount and displays successful transaction message

To Do

Improve Bug

Description

Steps to Reproduce:

1. Login to ParaBank application.

2. Navigate to Bill Pay.

3. Enter valid payee details.

4. Enter bill amount as 0.

5. Select source account.

6. Click Send Payment.

Expected Result:

System should validate the payment amount.

Bill amount should be greater than zero.

Appropriate validation message should be displayed.

Actual Result:

System processes the payment successfully.

"Bill Payment Complete" message is displayed.

Confirmation shows bill payment of \$0.00 was successful.

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System allows bill payment with negative amount and displays successful transaction message

To Do

Improve Bug

Description

Steps to Reproduce

1. Login to ParaBank application.

2. Navigate to Bill Pay.

3. Enter valid payee details.

4. Enter bill amount as -100.

5. Select source account.

6. Click Send Payment.

Expected Result

System should validate the bill amount.

Negative payment amounts should not be accepted.

Appropriate validation message should be displayed.

Actual Result

System processes the bill payment successfully.

"Bill Payment Complete" message is displayed.

Confirmation shows a payment amount of \$-100.00.

Attachments

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Test Case Analysis Report

Overview

The ParaBank Automation Framework was executed to validate the core banking functionalities of the application. Automated test cases were developed using Selenium WebDriver, Java, TestNG, Maven, and Jenkins.

Test Execution Summary

Metric	Count
Total Test Cases	18
Executed Test Cases	18
Passed Test Cases	14
Failed Test Cases	4
Blocked Test Cases	0

Module-wise Analysis

Module	Status
Registration	Passed
Login	Passed
Open Account	Passed
Account Overview	Passed
Transfer Funds	Passed with Defects Identified
Bill Pay	Passed with Defects Identified

Defect Analysis

Bug ID	Description	Severity
BUG-1	Same Account Transfer Allowed	Medium
BUG-2	Zero Amount Transfer Allowed	Medium
BUG-3	Zero Amount Bill Payment Allowed	Medium
BUG-4	Negative Amount Bill Payment Allowed	High

Test Coverage

- User Registration
- User Login
- Open New Account
- Account Overview
- Fund Transfer
- Bill Payment
- Cross Browser Execution (Chrome and Firefox)

Challenges Faced

- Browser compatibility configuration.
- Jenkins pipeline setup.
- Selenium WebDriver version compatibility.
- Dynamic test data management for user registration.

Conclusion

The automation framework successfully validated the major functionalities of the ParaBank application. Most test cases executed successfully, while four defects were identified during negative testing. The framework supports automated execution through Jenkins and generates execution reports using Extent Reports.

Project Summary

The ParaBank Automation Framework was developed using Selenium WebDriver, Java, TestNG, Maven, Jenkins, and GitHub following the Page Object Model (POM) design pattern. The framework automates key banking functionalities including User Registration, Login, Account Management, Fund Transfer, and Bill Payment.

A total of 18 automated test cases were executed. Out of these, 14 test cases passed successfully and 4 defects were identified during negative testing. The defects were documented and tracked using Jira. Jenkins was integrated for automated test execution, and Extent Reports were generated to provide detailed execution results.

The framework supports cross-browser testing on Chrome and Firefox and helps reduce manual testing effort through automation and reusable test components.

